

Problem 2.2

Given

$$\mathbf{A} = \begin{bmatrix} -1 & 3 \\ 4 & 2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 5 \\ -4 \\ 2 \end{bmatrix}.$$

(a) $5\mathbf{A}$

$$5\mathbf{A} = \begin{bmatrix} -5 & 15 \\ 20 & 10 \end{bmatrix}.$$

(b) $\mathbf{BA}$ $\mathbf{B}$ is 3×2 and $\mathbf{A}$ is 2×2 , so $\mathbf{BA}$ is 3×2 :

$$\mathbf{BA} = \begin{bmatrix} 4(-1) + (-3)(4) & 4(3) + (-3)(2) \\ 1(-1) + (-2)(4) & 1(3) + (-2)(2) \\ -2(-1) + 0(4) & -2(3) + 0(2) \end{bmatrix} = \begin{bmatrix} -16 & 6 \\ -9 & -1 \\ 2 & -6 \end{bmatrix}.$$

(c) $\mathbf{A}'\mathbf{B}'$ Since $\mathbf{A}'\mathbf{B}' = (\mathbf{BA})'$,

$$\mathbf{A}'\mathbf{B}' = \begin{bmatrix} -16 & -9 & 2 \\ 6 & -1 & -6 \end{bmatrix}.$$

(d) $\mathbf{C}'\mathbf{B}$ $\mathbf{C}' = [5, -4, 2]$ is 1×3 , $\mathbf{B}$ is 3×2 , so $\mathbf{C}'\mathbf{B}$ is 1×2 :

$$\mathbf{C}'\mathbf{B} = [5(4) + (-4)(1) + 2(-2), \quad 5(-3) + (-4)(-2) + 2(0)] = [12, -7].$$

(e) **Is $\mathbf{AB}$ defined?** $\mathbf{A}$ is 2×2 and $\mathbf{B}$ is 3×2 . For the product $\mathbf{AB}$ we would need the number of columns of $\mathbf{A}$ (which is 2) to equal the number of rows of $\mathbf{B}$ (which is 3). Since $2 \neq 3$, $\mathbf{AB}$ is **not defined**.

Problem 2.4

(a) $(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'$ Start from $\mathbf{AA}^{-1} = \mathbf{I}$. Taking transposes of both sides and using $' =$,

$$(\mathbf{AA}^{-1})' = \mathbf{I}' \implies (\mathbf{A}^{-1})'\mathbf{A}' = \mathbf{I}.$$

Similarly, from $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$,

$$(\mathbf{A}^{-1}\mathbf{A})' = \mathbf{I}' \implies \mathbf{A}'(\mathbf{A}^{-1})' = \mathbf{I}.$$

Thus $(\mathbf{A}^{-1})'$ acts as both a left and a right inverse of $\mathbf{A}'$. Since the inverse of a matrix is unique,

$$(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})'.$$

(b) $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ Compute

$$(\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{AB}) = \mathbf{B}^{-1}(\mathbf{A}^{-1}\mathbf{A})\mathbf{B} = \mathbf{B}^{-1}\mathbf{B} = \mathbf{I} = \mathbf{B}^{-1}\mathbf{B},$$

$$(\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{A}(\mathbf{BB}^{-1})\mathbf{A}^{-1} = \mathbf{AA}^{-1} = \mathbf{I} = \mathbf{AA}^{-1}.$$

Since $\mathbf{B}^{-1}\mathbf{A}^{-1}$ is both a left and right inverse of $\mathbf{AB}$, and inverses are unique,

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}.$$

Problem 2.8

Given $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix}$.

Eigenvalues

$$|\mathbf{A} - \lambda| = (1 - \lambda)(-2 - \lambda) - 4 = \lambda^2 + \lambda - 6 = 0 \implies (\lambda - 2)(\lambda + 3) = 0.$$

$$\lambda_1 = 2, \quad \lambda_2 = -3.$$

Eigenvectors For $\lambda_1 = 2$: solve $(\mathbf{A} - 2)v = 0$:

$$\begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} v = 0 \implies v_1 = 2v_2.$$

Taking $v = (2, 1)'$ and normalizing ($\|v\| = \sqrt{5}$):

$$e_1 = \left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right)'.$$

For $\lambda_2 = -3$: solve $(\mathbf{A} + 3)v = 0$:

$$\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} v = 0 \implies v_2 = -2v_1.$$

Taking $v = (1, -2)'$ and normalizing ($\|v\| = \sqrt{5}$):

$$e_2 = \left(\frac{1}{\sqrt{5}}, \frac{-2}{\sqrt{5}} \right)'.$$

Spectral decomposition

$$\mathbf{A} = \lambda_1 e_1 e_1' + \lambda_2 e_2 e_2' = 2 \cdot \frac{1}{5} \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} + (-3) \cdot \frac{1}{5} \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix}.$$

Numerically,

$$\mathbf{A} = \frac{1}{5} \begin{bmatrix} 8 & 4 \\ 4 & 2 \end{bmatrix} + \frac{1}{5} \begin{bmatrix} -3 & 6 \\ 6 & -12 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 5 & 10 \\ 10 & -10 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & -2 \end{bmatrix},$$

which recovers $\mathbf{A}$, as required.

Problem 2.11

Let $\mathbf{A}$ be $p \times p$ diagonal, with $a_{ij} = 0$ for $i \neq j$. We show $|\mathbf{A}| = a_{11}a_{22} \cdots a_{pp}$ by induction on p .

Base case ($p = 1$): $\mathbf{A} = [a_{11}]$, so $|\mathbf{A}| = a_{11}$, trivially true.

Inductive step: Assume the result holds for all diagonal matrices of size $(p-1) \times (p-1)$. By cofactor expansion along the first row (Definition 2A.24),

$$|\mathbf{A}| = a_{11}A_{11} + a_{12}A_{12} + \cdots + a_{1p}A_{1p}.$$

Since $\mathbf{A}$ is diagonal, $a_{12} = a_{13} = \cdots = a_{1p} = 0$, so

$$|\mathbf{A}| = a_{11}A_{11},$$

where $A_{11} = (-1)^{1+1}M_{11} = M_{11}$, and M_{11} is the determinant of the submatrix obtained by deleting the first row and first column of $\mathbf{A}$. This submatrix is itself a $(p-1) \times (p-1)$ diagonal matrix with diagonal entries $a_{22}, a_{33}, \dots, a_{pp}$. By the induction hypothesis,

$$M_{11} = a_{22}a_{33} \cdots a_{pp}.$$

Therefore,

$$|\mathbf{A}| = a_{11}(a_{22}a_{33} \cdots a_{pp}) = a_{11}a_{22} \cdots a_{pp}.$$

By induction, this holds for all $p \geq 1$. ■

Problem 2.24

Given

$$\mathbf{\Sigma} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(a) $\mathbf{\Sigma}^{-1}$ Since $\mathbf{\Sigma}$ is diagonal,

$$\mathbf{\Sigma}^{-1} = \begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/9 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(b) **Eigenvalues and eigenvectors of $\mathbf{\Sigma}$** Because $\mathbf{\Sigma}$ is diagonal, its eigenvalues are the diagonal entries, with the standard basis vectors as corresponding eigenvectors. Ordered from largest to smallest:

$$\lambda_1 = 9, \quad e_1 = (0, 1, 0)'; \quad \lambda_2 = 4, \quad e_2 = (1, 0, 0)'; \quad \lambda_3 = 1, \quad e_3 = (0, 0, 1)'.$$

(c) **Eigenvalues and eigenvectors of $\mathbf{\Sigma}^{-1}$** $\mathbf{\Sigma}^{-1}$ shares the same eigenvectors as $\mathbf{\Sigma}$, with eigenvalues $1/\lambda_i$. Ordered from largest to smallest:

$$\lambda_1^* = 1, \quad e_1 = (0, 0, 1)'; \quad \lambda_2^* = \frac{1}{4}, \quad e_2 = (1, 0, 0)'; \quad \lambda_3^* = \frac{1}{9}, \quad e_3 = (0, 1, 0)'.$$

Problem 2.30

Given $\boldsymbol{\mu}'_X = [4, 3, 2, 1]$,

$$\mathbf{\Sigma}_X = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 1 & 1 & 0 \\ 2 & 1 & 9 & -2 \\ 2 & 0 & -2 & 4 \end{bmatrix}, \quad \mathbf{X}^{(1)} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}, \quad \mathbf{X}^{(2)} = \begin{bmatrix} X_3 \\ X_4 \end{bmatrix},$$

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}.$$

(a) $E(\mathbf{X}^{(1)})$

$$E(\mathbf{X}^{(1)}) = \begin{bmatrix} 4 \\ 3 \end{bmatrix}.$$

(b) $E(\mathbf{A}\mathbf{X}^{(1)})$

$$E(\mathbf{A}\mathbf{X}^{(1)}) = \mathbf{A} E(\mathbf{X}^{(1)}) = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = 4 + 6 = 10.$$

(c) $\text{Cov}(\mathbf{X}^{(1)})$ Taking the rows/columns corresponding to X_1, X_2 from $\mathbf{\Sigma}_X$:

$$\text{Cov}(\mathbf{X}^{(1)}) = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

(d) $\text{Cov}(\mathbf{A}\mathbf{X}^{(1)})$

$$\text{Cov}(\mathbf{A}\mathbf{X}^{(1)}) = \mathbf{A} \text{Cov}(\mathbf{X}^{(1)}) \mathbf{A}' = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 3 + 4 = 7.$$

(e) $E(\mathbf{X}^{(2)})$

$$E(\mathbf{X}^{(2)}) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

(f) $E(\mathbf{B}\mathbf{X}^{(2)})$

$$E(\mathbf{B}\mathbf{X}^{(2)}) = \mathbf{B} E(\mathbf{X}^{(2)}) = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 - 2 \\ 4 - 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

(g) $\text{Cov}(\mathbf{X}^{(2)})$ Taking the rows/columns corresponding to X_3, X_4 from $\mathbf{\Sigma}_X$:

$$\text{Cov}(\mathbf{X}^{(2)}) = \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix}.$$

(h) $\text{Cov}(\mathbf{B}\mathbf{X}^{(2)})$

$$\text{Cov}(\mathbf{B}\mathbf{X}^{(2)}) = \mathbf{B} \text{Cov}(\mathbf{X}^{(2)}) \mathbf{B}'.$$

First,

$$\mathbf{B} \text{Cov}(\mathbf{X}^{(2)}) = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 9 & -2 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 13 & -10 \\ 20 & -8 \end{bmatrix}.$$

Then,

$$\text{Cov}(\mathbf{B}\mathbf{X}^{(2)}) = \begin{bmatrix} 13 & -10 \\ 20 & -8 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} 33 & 36 \\ 36 & 48 \end{bmatrix}.$$

(i) $\text{Cov}(\mathbf{X}^{(1)}, \mathbf{X}^{(2)})$ Taking rows 1, 2 and columns 3, 4 of $\mathbf{\Sigma}_X$:

$$\text{Cov}(\mathbf{X}^{(1)}, \mathbf{X}^{(2)}) = \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix}.$$

(j) $\text{Cov}(\mathbf{A}\mathbf{X}^{(1)}, \mathbf{B}\mathbf{X}^{(2)})$

$$\text{Cov}(\mathbf{A}\mathbf{X}^{(1)}, \mathbf{B}\mathbf{X}^{(2)}) = \mathbf{A} \text{Cov}(\mathbf{X}^{(1)}, \mathbf{X}^{(2)}) \mathbf{B}'.$$

First,

$$\mathbf{A} \text{Cov}(\mathbf{X}^{(1)}, \mathbf{X}^{(2)}) = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 2 \end{bmatrix}.$$

Then,

$$\begin{bmatrix} 4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} = \begin{bmatrix} 4 - 4 & 8 - 2 \end{bmatrix} = \begin{bmatrix} 0 & 6 \end{bmatrix}.$$

So

$$\text{Cov}(\mathbf{A}\mathbf{X}^{(1)}, \mathbf{B}\mathbf{X}^{(2)}) = \begin{bmatrix} 0 & 6 \end{bmatrix}.$$

Problem 2.41

Given $\boldsymbol{\mu}'_X = [3, 2, -2, 0]$, $\boldsymbol{\Sigma}_X = 3\mathbf{I}_4$, and

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{bmatrix}.$$

(a) $E(\mathbf{A}\mathbf{X})$

$$E(\mathbf{A}\mathbf{X}) = \mathbf{A}\boldsymbol{\mu}_X = \begin{bmatrix} 1(3) + (-1)(2) + 0(-2) + 0(0) \\ 1(3) + 1(2) + (-2)(-2) + 0(0) \\ 1(3) + 1(2) + 1(-2) + (-3)(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 9 \\ 3 \end{bmatrix}.$$

(b) $\text{Cov}(\mathbf{A}\mathbf{X})$ Since $\boldsymbol{\Sigma}_X = 3\mathbf{I}$,

$$\text{Cov}(\mathbf{A}\mathbf{X}) = \mathbf{A}\boldsymbol{\Sigma}_X\mathbf{A}' = 3\mathbf{A}\mathbf{A}'.$$

Computing $\mathbf{A}\mathbf{A}'$ entrywise (dot products of the rows of $\mathbf{A}$):

$$\mathbf{A}\mathbf{A}' = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 12 \end{bmatrix}.$$

Therefore,

$$\text{Cov}(\mathbf{A}\mathbf{X}) = 3 \begin{bmatrix} 2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 12 \end{bmatrix} = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 36 \end{bmatrix}.$$

(c) **Which pairs have zero covariance?** Let Y_1, Y_2, Y_3 denote the three linear combinations given by the rows of $\mathbf{A}$ (so $\mathbf{A}\mathbf{X} = (Y_1, Y_2, Y_3)'$). Since $\text{Cov}(\mathbf{A}\mathbf{X})$ is diagonal, every off-diagonal covariance is zero. Hence **all three pairs** (Y_1, Y_2) , (Y_1, Y_3) , and (Y_2, Y_3) have zero covariance – the three linear combinations are mutually uncorrelated.